1. Find a general solution to the equation $y' = x^2/y^3$.

Solution.

$$\begin{aligned} \frac{dy}{dx} &= x^2/y^3 \\ y^3\,dy &= x^2\,dx \\ y^4/4 &= x^3/3 + C \\ y^4 &= 4x^3/3 + C \\ y &= \sqrt[4]{4x^3/3 + C}. \end{aligned}$$

2. Consider the sequence $(a_n) = \left(2, \frac{3}{4}, \frac{4}{9}, \frac{5}{16}, \frac{6}{25}, \dots\right)$. Find a formula for the nth term a_n. Compute $\lim_{n\to\infty} a_n$.

Solution. We have $a_n = \frac{n+1}{n^2}$, and thus

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n+1}{n^2} = 0$$

since the denominator has larger degree.

3. Compute $\displaystyle\sum_{n=1}^{\infty} \frac{5^{n-1}}{2^n}$.

Solution. This is a geometric series with initial term $a = 1/2$ and common ratio $r = 5/2$. Since $|r| > 1$, the series diverges, that is, $\sum_{n=1}^{\infty} \frac{5^{n-1}}{2^n} = \infty$.

4. Does the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+4}$ converge or diverge?

Solution. Either one of the comparison tests is probably easiest here.

We see that $\frac{1}{n^2+4} < \frac{1}{n^2}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (by the p-series test), we can conclude by the direct comparison test that so does $\sum_{n=1}^{\infty} \frac{1}{n^2+4}$.

We could also use the limit comparison test: we have

$$\lim_{n\to\infty} \frac{\frac{1}{n^2+4}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^2}{n^2+4} = 1.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (by the p-series test), so does $\sum_{n=1}^{\infty} \frac{1}{n^2+4}$.

Alternatively, we can work this out with the integral test. We have

$$\begin{aligned}\int_1^\infty \frac{1}{x^2+4}\,dx &= \lim_{t\to\infty}\int_1^t \frac{1}{4}\frac{1}{1+(x/2)^2}\,dx\\ &= \lim_{t\to\infty}\frac{1}{2}\arctan(x/2)\Big|_1^t\\ &= \lim_{t\to\infty}\arctan(t/2)-\arctan(1/2) = \pi/2-\arctan(1/2) < \infty.\end{aligned}$$

Since this integral converges, the series must also converge by the integral test.